

6/20

Test Assessment -1

1. Reverse a string using slicing

Ans: string = "Hello World"

reverse = ""

for ch in string:

reverse = ch + reverse

print(reverse)

String = "Hello World"

reverse = string[::-1]

print(reverse)

2. Count words in a sentence:

Ans: sentence = "A Quick Brown Fox Jump Over the Lazy Dog"

words = sentence.split()

print("Number of words:", len(words))

3. Remove duplicate characters

Ans: string = "carrot"

result = ""

for ch in string:

if ch not in result:

result += ch

print(result)

4. Difference between lists & sets

lists

sets

ordered collection

Unordered collection

Allow duplicate values

Does not allow duplicate values

Uses []

Uses {}

Elements can be accessed by index

No indexing

5. Can sets contain lists? If yes How? If no why?

Ans: No, because list elements are mutable (can be modified after creation) while set elements must be immutable (hashable).

6. Why sets are faster

Ans

7. ~~remove~~: Difference between `remove()` & `discard()`

Ans: `remove`: Removes the specified element but raises `KeyError` if element not found.
`discard`: Removes the specified element but does not raise any error if element not found.

8. How many methods to insert elements in the list

Ans There are 3 methods:

- i] `append(x)`: adds element at the end
- ii] `extend(iterable)`: adds multiple elements
- iii] `insert(index, x)`: inserts element at specified position.

9. How many methods to delete elements in lists?

Ans There are 5 methods:

- `remove(x)`: removes first occurrence of specified value
- `pop(n)`: Removes element at given index
- `pop()`: Removes last element
- `clear()`: Removes all elements
- `del()`: deletes an element

10. How many operators in python?

Ans. There are 7 main types of operators:

- ✓ Arithmetic : $+$, $-$, $*$, $/$, $//$, $**$, $\%$
- ✓ Relational : $==$, $!=$, $>$, $<$, $>=$, $<=$
- ✓ Assignment : $=$, $+=$, $-=$, $*=$, $/=$, $//=$, $\%=$, $**=$
- ✓ Logical : and , or , not
- ✓ Bitwise : $\&$, $|$, $\wedge$, $\sim$, $<<$, $>>$
- ✓ membership : in , $not\ in$
- ✓ Identity : is , $is\ not$

11. Write precedence Rule of arithmetic in Python

Ans. High : $()$

Medium : $*$, $/$, $//$, $**$, $\%$

low : $+$, $-$

12. How to sort elements in list using slicing

Ans:

13. What is Dictionary?

Ans. A dictionary is a mutable collection that stores data in key value pairs.

14. How many methods to create dictionary

Ans. There are 3 common methods:

i] Using curly braces : $\{ \}$

ii] Using $dict()$

iii] Empty Dictionary

15] How many methods to access value in dictionary

→ Methods used to access values in dictionary:

- i] Using Key : `d["name"]`
- ii] Using items() : `d.items()`

16] How many methods to delete elements in dictionary?

→ Methods to delete elements are:

- i] `pop(Key)`
- ii] `popitem()`
- iii] `clear()`
- iv] `del d[Key]`

17]

→ Union, Intersection, Difference, Symmetric Difference

18] What is packing variable in tuple?

→ Packing means storing multiple values into a single tuple.

eg: `t = 10, 20, 30`
`print(t)`

19] What is unpacking in tuple?

→ Unpacking means assigning tuple elements to separate variables.

20]

Set

stores only values

No duplicate values

No indexing

Mutable

Dictionary

Stores Key-value pairs

Keys must be unique

(Values may repeat)

Access values using Keys

Mutable